# How BARCS Finds Important Genes

A student guide in two chapters, with worked examples and exercises

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BARCS version 0.2.1 · Reading level: grades 7 to 9

**About this guide.** This guide explains how the BARCS computer program works. You need only arithmetic, fractions, percents, powers and square roots. The guide teaches logarithms when it uses them. Both chapters use the same small set of numbers, so you can follow every step. The guide uses the writing rules of ASD-STE100 Simplified Technical English.

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# Chapter 1

## The ideas

Counts, proportions, noise and the design matrix

# The experiment: a CRISPR screen

Each cell in your body has about 20,000 genes. A gene is a set of instructions. Scientists want to know which genes a cell needs, for example to survive a cancer drug.

A **CRISPR screen** tests thousands of genes at the same time. It works in four steps:

1. Scientists make a **library** of short pieces of RNA, called **guides**. Each guide switches off one gene.
2. They put one guide into each cell. Now each cell has one gene switched off.
3. They grow the cells. Some cells grow under normal conditions (**control**). Other cells grow with a drug (**drug**).
4. A sequencing machine reads the guides in each group. The machine counts how many times it reads each guide. This number is a **read count**.

If cells need a gene to survive the drug, the cells that lose that gene die. Their guide becomes rare in the drug group. So the guide count goes down.

**Remember**

BARCS does not look at cells directly. It looks at read counts. A guide that becomes rare tells us that its gene is important.

Scientists repeat each group more than one time. Each repeat is a **replicate**. Each sample that the machine reads is a **library**. In this guide, we use four libraries: two control replicates and two drug replicates.

# The data

The machine gives two numbers for each library:

- the read count of our guide, and
- the total read count of all guides in that library.

Table 1 shows the data for one guide. We use these numbers in all sections of this guide.

**What the symbols mean**

| | |
|---|---|
| $K$ | The read count of one guide in one library. Library 1 has $K = 180$. |
| $N$ | The total read count of all guides in one library. Library 1 has $N = 1{,}000{,}000$. |
| $i$ | The library number (1, 2, 3 or 4). We write $K_i$ for the count in library $i$. So $K_2 = 240$. |

Table 1: Read counts for one guide in four libraries.

| Library | Group | Guide count $K$ | Total count $N$ |
|---|---|---|---|
| 1 | control | 180 | 1,000,000 |
| 2 | control | 240 | 1,200,000 |
| 3 | drug | 120 | 1,000,000 |
| 4 | drug | 80 | 800,000 |

**Remember**

Always record $N$ before you remove any guides from the table. If you remove guides first, the totals become wrong. Then all the later numbers become wrong too.

## Step 1: Turn counts into proportions

Library 2 has more reads than library 1 because the machine read it more. So we cannot compare raw counts. We divide each count by its library total. The result is a **proportion**:

$$p = \frac{K}{N}.$$

Proportions are very small numbers. So we multiply by one million. The result is the count **per million** reads.

**Worked example 1: proportions**

Library 1: $p = 180 \div 1{,}000{,}000 = 0.000180 \quad = 180$ per million
Library 2: $p = 240 \div 1{,}200{,}000 = 0.000200 \quad = 200$ per million
Library 3: $p = 120 \div 1{,}000{,}000 = 0.000120 \quad = 120$ per million
Library 4: $p = \ 80 \div \ 800{,}000 = 0.000100 \quad = 100$ per million

The two control libraries give 180 and 200 per million. The two drug libraries give 120 and 100 per million. The guide is less common with the drug.

**What the symbols mean**

$p$ The proportion of all reads that belong to the guide. It is always between 0 and 1.

## Step 2: Compare the groups with a fold change

We want one number that tells how much the guide changes. First, we combine the replicates of each group. We add their counts and add their totals:

$$p_{\text{control}} = \frac{180 + 240}{1{,}000{,}000 + 1{,}200{,}000} = \frac{420}{2{,}200{,}000} = 0.000191 \text{ (191 per million)},$$

$$p_{\text{drug}} = \frac{120 + 80}{1{,}000{,}000 + 800{,}000} = \frac{200}{1{,}800{,}000} = 0.000111 \text{ (111 per million)}.$$

The **fold change** is the drug proportion divided by the control proportion:

$$\text{fold change} = \frac{p_{\text{drug}}}{p_{\text{control}}} = \frac{0.000111}{0.000191} = 0.58.$$

A fold change of 0.58 means that the guide keeps only 58% of its control level. It loses 42%.

**Remember**

- Fold change $= 1$: no change.
- Fold change $< 1$: the guide becomes rarer. Cells may need this gene.
- Fold change $> 1$: the guide becomes more common. Losing this gene may help the cells.

## Logarithms make changes symmetric

A fold change of 2 means "double". A fold change of 0.5 means "half". These changes have the same size, but the numbers 2 and 0.5 do not look equal. A **logarithm** fixes this problem.

The **base-2 logarithm**, $\log_2$, tells how many times you double.

| Fold change | 0.25 | 0.5 | 1 | 2 | 4 |
|---|---|---|---|---|---|
| $\log_2$(fold change) | $-2$ | $-1$ | 0 | 1 | 2 |

Now "double" is $+1$ and "half" is $-1$. The sizes look equal.

**Worked example 2: log fold change**

Our fold change is 0.58. It is between 0.5 and 1. So $\log_2(0.58)$ is between $-1$ and 0.
A calculator gives $\log_2(0.58) = -0.78$.
The guide drops by 0.78 "halvings".

BARCS uses the **natural logarithm**, ln. It counts powers of the number $e = 2.718\ldots$ instead of powers of 2. The idea is the same. To change back from ln, you raise $e$ to that power: if $\ln(x) = b$, then $x = e^b$. For our guide, $\ln(0.58) = -0.54$.

## Step 3: Odds and log-odds

BARCS works with **odds** instead of proportions. The odds compare the reads that belong to the guide with the reads that do not:

$$\text{odds} = \frac{p}{1-p}.$$

The **log-odds** is the natural logarithm of the odds:

$$\text{log-odds} = \ln\left(\frac{p}{1-p}\right).$$

**Worked example 3: odds and log-odds**

Control: $p = 0.000191$. So odds $= 0.000191 \div 0.999809 = 0.000191$.
The odds and the proportion are almost equal. This is true for all small proportions, because $1 - p$ is almost 1.
Log-odds for control: $\ln(0.000191) = -8.56$.
Log-odds for drug: $\ln(0.000111) = -9.10$.
Difference: $-9.10 - (-8.56) = -0.54$.
The difference of the log-odds is $-0.54$. This is the same as ln(fold change) from Step 2.

**Remember**

For guides, a difference in log-odds is almost exactly the natural log of the fold change. So you can read BARCS results as log fold changes.

Why use log-odds at all? A proportion must stay between 0 and 1. A log-odds can be any number, positive or negative. This property lets BARCS add effects together safely, as Step 5 shows.

## Step 4: Two kinds of noise

The two control libraries gave 180 and 200 per million. They are not equal. Replicates always differ a little. We call this difference **noise**. BARCS must know how big the noise is. Then it can tell a real change from a chance change.

Guide counts have two kinds of noise:

1. **Counting noise.** The machine reads a random sample of all the guides. Think of a bag with a million marbles. You take out a handful. You do not get exactly the same number of red marbles each time.
2. **Biological noise.** Each replicate is a different group of living cells. The cells grow a little differently each time.

BARCS measures the size of the noise with the **variance**. The variance is the average squared distance of a value from its expected value. The **standard deviation** is the square root of the variance. It is the typical distance in the same units as the counts.

This step builds the variance formula of BARCS in four parts. In all formulas, the sign $\times$ means "multiply". So $N \times \mu$ means "$N$ multiplied by $\mu$".

**What the symbols mean**

| | |
|---|---|
| $N$ | The library total, as before. Our example uses $N = 1{,}000{,}000$. |
| $\mu$ | "mu". The proportion of the guide that we expect in the library. For the control group, our best value is $\mu = 0.000191$. It is like $p$ from Step 1, but it is the expected value, not the value that we measure. |
| $\rho$ | "rho". A number that measures the biological noise. If $\rho = 0$, only counting noise exists. A bigger $\rho$ means more biological noise. Part 3 explains it. |

## Part 1: The expected count

The machine reads $N$ reads. Each read belongs to our guide with the chance $\mu$. So we expect

$$\text{expected count} = N \times \mu.$$

**Worked example 4a: expected count**

$N \times \mu = 1{,}000{,}000 \times 0.000191 = 191$ reads.
The real counts are near 191, but not exactly 191. The variance tells how far they normally are from 191.

## Part 2: Counting noise

Start with **one read**. It belongs to our guide (value 1) or it does not (value 0). We need the variance of this one value. We do it in three steps: first with five marbles, then the reason for the squares, then the general formula.

**Step A: Five marbles.** Put five marbles in a bag: one red and four blue. Give the red marble the value 1 and each blue marble the value 0. So the chance of red is $\mu = 1/5 = 0.2$. The average value is $(1 + 0 + 0 + 0 + 0)/5 = 0.2$. This average is $\mu$.

Now find the distance of each marble from the average:

| Marble | Value | Distance (value − 0.2) | Squared distance |
|---|---|---|---|
| red | 1 | +0.8 | 0.64 |
| blue | 0 | −0.2 | 0.04 |
| blue | 0 | −0.2 | 0.04 |
| blue | 0 | −0.2 | 0.04 |
| blue | 0 | −0.2 | 0.04 |
| Sum | | $0.8 - 0.8 = 0$ | 0.80 |
| Average (sum ÷ 5) | | 0 | **0.16** |

Look at the distance column. The positive and the negative distances cancel. Their sum is 0. This is always true: the distances from an average always add up to 0. So the plain average distance cannot measure the spread. It is 0 for every bag.

To stop the cancelling, we square each distance. A square is never negative. The average squared distance is $0.80/5 = 0.16$. This is the **variance** of one marble.

**Step B: Why squares, and not only the size of the distance?** We could also drop the minus signs and use the size of each distance. But BARCS needs a measure that **adds up**: the spread of a sum of independent values must be the sum of their spreads. Part 2 adds up a million reads, and Part 4 adds two kinds of noise. Squares have this property. Sizes do not.

**Worked example 4b: two coins**

Throw two fair coins. Heads = 1, tails = 0. Each coin has $\mu = 0.5$. The four results are equally likely.

| Coins | Total | Distance from 1 | Squared distance |
|---|---|---|---|
| tails, tails | 0 | −1 | 1 |
| tails, heads | 1 | 0 | 0 |
| heads, tails | 1 | 0 | 0 |
| heads, heads | 2 | +1 | 1 |

- **Variance of the total:** $(1 + 0 + 0 + 1)/4 = 0.5$.
- **Variance of one coin:** distances are $\pm 0.5$, squared 0.25. Two coins: $0.25 + 0.25 = 0.5$. **The same.** Squares add up.
- **Size of distance, total:** $(1 + 0 + 0 + 1)/4 = 0.5$.
- **Size of distance, one coin:** 0.5. Two coins: $0.5 + 0.5 = 1$. **Not the same** as 0.5. Sizes do not add up.

The statistician Ronald Fisher gave the name *variance* to the average squared distance in

1918. He chose it because the variances of independent values add up.[1] The idea of squared distances is older. In 1805 Adrien-Marie Legendre fitted lines by making the sum of squared distances as small as possible ("least squares"), and Carl Friedrich Gauss used the same idea.[2] BARCS still uses least squares in Chapter 2.

**Step C: The general formula.** Now replace "one marble in five" by any chance $\mu$. The table has only two kinds of rows:

| Value | How often (chance) | Distance from $\mu$ | Squared distance |
|---|---|---|---|
| 1 | $\mu$ | $1-\mu$ | $(1-\mu)^2$ |
| 0 | $1-\mu$ | $-\mu$ | $\mu^2$ |

The average squared distance is each squared distance multiplied by how often it happens, then added:

$$\text{variance of one read} = \mu \times (1-\mu)^2 + (1-\mu) \times \mu^2.$$

In the five-marble bag: $0.2 \times 0.8^2 + 0.8 \times 0.2^2 = 0.128 + 0.032 = 0.16$. This is the same 0.16 as in the table. (The table gave $0.64 \times \frac{1}{5} + 0.04 \times \frac{4}{5}$.)

**Step D: From the long formula to $\mu \times (1-\mu)$.** We now open every bracket, one step at a time. The right column does the same step with $\mu = 0.2$, so you can check each line.

**First, open the square.** A square means "multiply by itself": $(1-\mu)^2 = (1-\mu) \times (1-\mu)$. Multiply each part of the first bracket by each part of the second bracket:

$$(1-\mu) \times (1-\mu) = 1 \times 1 \;-\; 1 \times \mu \;-\; \mu \times 1 \;+\; \mu \times \mu = 1 - 2 \times \mu + \mu^2.$$

Check: $0.8 \times 0.8 = 0.64$, and $1 - 2 \times 0.2 + 0.04 = 0.64$.

[1] R. A. Fisher (1918). The correlation between relatives on the supposition of Mendelian inheritance. *Transactions of the Royal Society of Edinburgh* 52: 399–433.

[2] S. M. Stigler (1986). *The History of Statistics.* Harvard University Press, chapter 1.

**Then, expand everything.**

| **With letters** | **With $\mu = 0.2$** |
|---|---|
| 1. $\mu \times (1-\mu)^2 + (1-\mu) \times \mu^2$ | $0.2 \times 0.64 + 0.8 \times 0.04$ |
| The formula from Step C. | |
| 2. $\mu \times (1 - 2 \times \mu + \mu^2) + (1-\mu) \times \mu^2$ | $0.2 \times 0.64 + 0.8 \times 0.04$ |
| Put in the opened square. | |
| 3. $(\mu - 2 \times \mu^2 + \mu^3) + (\mu^2 - \mu^3)$ | $(0.2 - 0.08 + 0.008) + (0.04 - 0.008)$ |
| Multiply $\mu$ into the first bracket and $\mu^2$ into the second. | |
| 4. $\mu - 2 \times \mu^2 + \mu^2 + \mu^3 - \mu^3$ | $0.2 - 0.08 + 0.04 + 0.008 - 0.008$ |
| Put the same powers next to each other. | |
| 5. $\mu - \mu^2$ | $0.2 - 0.04 = 0.16$ |
| $-2 \times \mu^2 + \mu^2 = -\mu^2$, and $+\mu^3 - \mu^3 = 0$. | |
| 6. $\mu \times (1-\mu)$ | $0.2 \times 0.8 = 0.16$ |
| Take $\mu$ out: $\mu - \mu \times \mu = \mu \times (1-\mu)$. | |

Every line gives 0.16. So the long formula and the short formula are always equal:

$$\mu \times (1-\mu)^2 + (1-\mu) \times \mu^2 = \mu \times (1-\mu).$$

**A shorter way.** Both parts of the long formula contain $\mu \times (1-\mu)$. Take this common part out:

$$\mu \times (1-\mu) \times \Big[(1-\mu) + \mu\Big] = \mu \times (1-\mu) \times 1 = \mu \times (1-\mu).$$

The bracket is 1 because $(1-\mu) + \mu = 1$. Check: $0.2 \times 0.8 \times (0.8 + 0.2) = 0.16$.

**Worked example 4c: 100 reads**

**100 reads:** The reads are independent. So their variances add up (Step B). 100 reads with $\mu = 0.2$ have the variance $100 \times 0.16 = 16$. The expected count is $100 \times 0.2 = 20$. The standard deviation is $\sqrt{16} = 4$.
**Check by repeating.** We took 100 marbles ten times. The red counts were 17, 19, 21, 25, 17, 25, 27, 22, 21 and 14. They are near 20, and they usually differ from 20 by about 4.

For $N$ reads, the variances of the $N$ reads add up. So the counting noise is

$$\text{counting noise} = N \times \mu \times (1-\mu).$$

For our guide: $1{,}000{,}000 \times 0.000191 \times 0.999809 = 191$.

## Part 3: Biological noise — the proportion wobbles

Counting noise assumes that every replicate has exactly the same true proportion $\mu$. But each replicate is a different group of living cells. Some cells grow a little faster, and some die by chance. So the true proportion of the guide is a little different in each replicate.

**What is the wobble?** We write $p$ for the true proportion of the guide in one replicate. (This is the proportion in the cells, before the machine reads anything.) Over many replicates, the average of $p$ is $\mu$. The **wobble** is how far $p$ moves away from $\mu$ from one replicate to the next. We measure it with the same rule as every variance: the average squared distance. We call it $V$:

$$V = \text{average of } (p - \mu)^2.$$

Figure 1 shows four replicates. The true proportion is 0.1 in two of them and 0.3 in the other two. The average is $\mu = 0.2$. Each distance is 0.1, each squared distance is 0.01, so the wobble is $V = 0.01$. If every replicate had $p = 0.2$, all the arrows would have length 0, and the wobble would be $V = 0$.

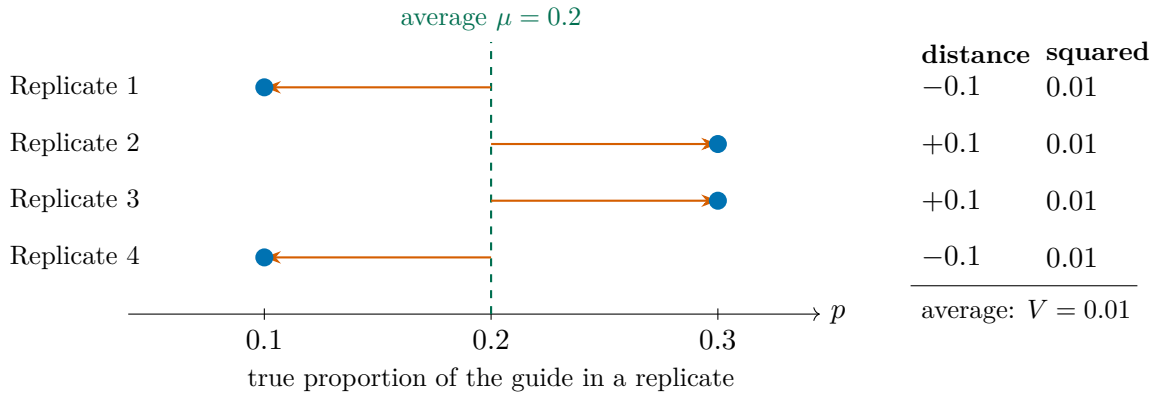



Figure 1: The wobble. Each dot is the true proportion $p$ in one replicate. Each arrow is its distance from the average $\mu$. The wobble $V$ is the average of the squared distances.

**Where $\rho$ comes from: two reads from the same replicate.** The machine reads many reads from one replicate. Take two of them. Ask this question: *if the first read is our guide, is the second read more likely to be our guide too?*

- **Without a wobble: no.** Every replicate has $p = \mu$. Each read is our guide with the chance $\mu$, whatever the other read is. The chance that both reads are our guide is $\mu \times \mu$.
- **With a wobble: yes.** If the first read is our guide, this replicate probably has a high $p$. So the second read is also more likely to be our guide. Reads from the same replicate **move together**.

Scientists measure how much two values move together with the **correlation**. A correlation of 0 means that the two values are unrelated. A correlation of 1 means that they are always the same. The number $\rho$ is exactly this: **the correlation between two reads from the same replicate**. That is also why it has the name $\rho$: it is the usual letter for a correlation.[3]

Now we calculate it. We do it in three steps.

**Step 1: the chance that both reads are our guide.** First look inside *one* replicate, with the true proportion $p$. Each read is our guide with the chance $p$. When $p$ is fixed, the two reads do not affect each other, like two throws of a coin. So the chance that both are our guide is $p \times p$. For example, with $p = 0.3$ it is $0.3 \times 0.3 = 0.09$.

Now look at *all* replicates. Each replicate has its own $p$. So the chance that both reads are our guide is the average of $p \times p$ over the replicates. In case B, half of the replicates give $0.1 \times 0.1 = 0.01$ and half give $0.3 \times 0.3 = 0.09$. The average is $(0.01 + 0.09)/2 = 0.05$. Without a wobble, it would be $0.2 \times 0.2 = 0.04$. Table 2 shows the three cases of Figure 2.

| Case ($\mu = 0.2$) | Both reads are our guide | If there were no wobble ($\mu \times \mu$) | Extra (= covariance) | Correlation $\rho$ = extra $\div$ 0.16 |
|---|---|---|---|---|
| A. $p = 0.2$ always | $0.2 \times 0.2 = 0.04$ | 0.04 | 0 | 0 |
| B. $p = 0.1$ or 0.3 | $(0.01 + 0.09)/2 = 0.05$ | 0.04 | 0.01 | 0.0625 |
| C. $p = 0$ or 1 | $0.8 \times 0 + 0.2 \times 1 = 0.2$ | 0.04 | 0.16 | 1 |

Table 2: Two reads from the same replicate. The wobble makes "both reads are our guide" more likely than $\mu \times \mu$. The extra is the covariance. It is always equal to the wobble $V$.

**Step 2: the extra is the covariance, and it equals $V$.** The extra over $\mu \times \mu$ shows how much the two reads move together. Its name is the **covariance**. It is the average of (distance of read 1 from $\mu$) $\times$ (distance of read 2 from $\mu$). Call the two reads $X_1$ and $X_2$ (each is 1 or 0). Open the brackets:

$$\begin{aligned}
\text{covariance} &= \text{average of } (X_1 - \mu) \times (X_2 - \mu) \\
&= \text{average of } \Big(X_1 \times X_2 - \mu \times X_2 - X_1 \times \mu + \mu \times \mu\Big) \\
&= \text{average of } X_1 \times X_2 - \mu \times \mu - \mu \times \mu + \mu \times \mu \\
&= (\text{chance that both are our guide}) - \mu \times \mu.
\end{aligned}$$

The third line uses that the average of each read is $\mu$. The last line needs one more fact. The product $X_1 \times X_2$ has only four possible results:

[3] In statistics, $\rho$ is called the *intra-class correlation*: the correlation of two trials in the same group. The beta-binomial model, in which the true proportion wobbles between groups, was described by J. G. Skellam (1948), *Journal of the Royal Statistical Society B* 10: 257–261.

| $X_1$ | $X_2$ | $X_1 \times X_2$ | |
|---|---|---|---|
| 1 | 1 | 1 | both reads are our guide |
| 1 | 0 | 0 | |
| 0 | 1 | 0 | |
| 0 | 0 | 0 | |

So the average of $X_1 \times X_2$ is $1\times$ (chance both are our guide) + $0\times$ (the other chances) = the chance that both are our guide. In case B: $0.05 - 0.04 - 0.04 + 0.04 = 0.05 - 0.04 = 0.01$. Now put in Step 1. The chance that both reads are our guide is the average of $p \times p$, which we write as the average of $p^2$. So

$$\text{covariance} = \text{average of } p^2 - \mu^2.$$

Case B: $0.05 - 0.04 = 0.01$.

Last, we show that the right side is the wobble $V$. Start from the definition of $V$ and open the square:

$$\begin{aligned}
V &= \text{average of } (p-\mu) \times (p-\mu) && \text{(definition of } V\text{)} \\
&= \text{average of } \left(p^2 - 2 \times \mu \times p + \mu^2\right) && \text{(multiply out)} \\
&= \text{average of } p^2 - 2 \times \mu \times (\text{average of } p) + \mu^2 && \text{(average each part)} \\
&= \text{average of } p^2 - 2 \times \mu^2 + \mu^2 && \text{(the average of } p \text{ is } \mu\text{)} \\
&= \text{average of } p^2 - \mu^2.
\end{aligned}$$

Case B: $0.05 - 2 \times 0.2 \times 0.2 + 0.04 = 0.05 - 0.08 + 0.04 = 0.01$.

The covariance and $V$ are both equal to the average of $p^2 - \mu^2$. So

$$\text{covariance of two reads} = V.$$

In case B both are 0.01. The wobble of the proportion *is* the covariance of two reads.

**Step 3: the correlation.** The correlation is the covariance divided by the standard deviations of the two values.[4] Each read has the variance $\mu \times (1-\mu)$ (Part 2), so each standard deviation is $\sqrt{\mu \times (1-\mu)}$. The product of the two standard deviations is $\mu \times (1-\mu)$. So

$$\rho = \frac{\text{covariance}}{\text{SD of read 1} \times \text{SD of read 2}} = \frac{V}{\mu \times (1-\mu)}.$$

Check with case B: $\rho = 0.01 \div 0.16 = 0.0625$. A computer drew 100,000 pairs of reads from case-B replicates and found the correlation 0.06. In case C, the two reads are always the same (both 0 or both 1), and the correlation is 1.

[4] This is Pearson's correlation coefficient: covariance ÷ (standard deviation of value 1 × standard deviation of value 2).

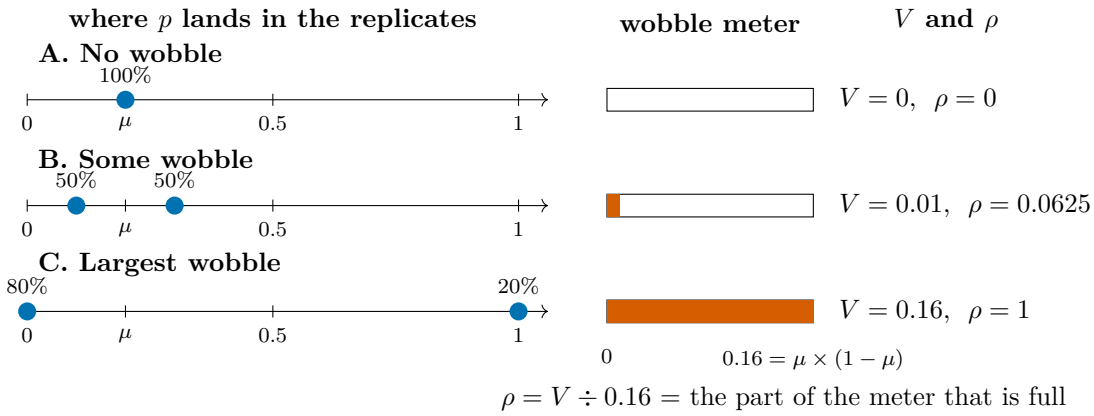



Figure 2: $\rho$, the correlation of two reads from the same replicate. It is 0 with no wobble and 1 with the largest possible wobble, $\mu \times (1 - \mu)$. In between, $\rho$ tells how full the "meter" is.

**Why $\rho$ is never more than 1.** A correlation cannot be more than 1. Here this means that the wobble can never be more than $\mu \times (1 - \mu)$. The table shows why, with checks for case B and case C.

| **Rule** | **Case B** ($p = 0.1, 0.3$) | **Case C** ($p = 0, 1$) |
|---|---|---|
| a. $V$ = average of $p^2 - \mu^2$ | $0.05 - 0.04 = 0.01$ | $0.2 - 0.04 = 0.16$ |
| b. $p \times p \leq p$ for every $p$ from 0 to 1 | $0.01 \leq 0.1$, $0.09 \leq 0.3$ | $0 = 0$, $1 = 1$ |
| c. so: average of $p^2 \leq \mu$ | $0.05 \leq 0.2$ | $0.2 = 0.2$ |
| d. so: $V \leq \mu - \mu^2 = \mu \times (1 - \mu)$ | $0.01 \leq 0.16$ | $0.16 = 0.16$ |

- **Line a.** Open the square: $(p - \mu)^2 = p^2 - 2 \times \mu \times p + \mu^2$. Take the average. The average of $p$ is $\mu$, so the middle part becomes $2 \times \mu \times \mu$. The result is: average of $p^2 - 2 \times \mu^2 + \mu^2$ = average of $p^2 - \mu^2$.
- **Line b.** A number from 0 to 1 does not grow when you multiply it by itself: $0.3 \times 0.3 = 0.09$. It stays the same only for 0 and 1.
- **Line c.** Each $p^2$ is at most $p$. So the average of $p^2$ is at most the average of $p$, which is $\mu$.
- **Line d.** Put line c into line a.

**Turn the formula around.** Multiply both sides of the formula for $\rho$ by $\mu \times (1 - \mu)$:

$$\rho = \frac{V}{\mu \times (1 - \mu)} \quad \Longrightarrow \quad \text{covariance of two reads} = V = \rho \times \mu \times (1 - \mu).$$

Check with case B: $0.0625 \times 0.16 = 0.01$. We use this in Part 4.

**Worked example 4d: the wobble of our guide**

**Variance of one read:** $\mu \times (1 - \mu) = 0.000191 \times 0.999809 = 0.000191$.
**Correlation of two reads:** BARCS estimates $\rho = 0.000000386$ (Step 7 in Chapter 2 shows how). Two reads from the same library are almost unrelated.
**The wobble (= covariance):** $V = \rho \times \mu \times (1 - \mu) = 0.000000386 \times 0.000191 = 0.0000000000737$.
**In words:** the square root of $V$ is 0.0000086, which is 8.6 per million. So the true proportion of our guide moves by about 8.6 per million between replicates, for example from 191 to 183 or 199 per million.

## Part 4: Add up the reads — where $N \times (N - 1)$ comes from

The count $K$ is the number of reads that are our guide. So $K$ is a sum of $N$ values, one for each read: $K = X_1 + X_2 + \cdots + X_N$, where each $X$ is 1 (our guide) or 0 (not our guide).

**Two reads first.** Part 2 (Step B) said that the variances of *independent* values add up. But reads from the same replicate are not independent: they move together. Look at the distance of the sum from its average. It is the sum of the two distances, $d_1 + d_2$. Square it:

$$(d_1 + d_2)^2 = d_1 \times d_1 + d_2 \times d_2 + 2 \times d_1 \times d_2.$$

Take the average of each part. The average of $d_1 \times d_1$ is the variance of read 1. The average of $d_1 \times d_2$ is the covariance (Part 3, Step 2). So

$$\text{variance of } (X_1 + X_2) = \text{variance} + \text{variance} + 2 \times \text{covariance}.$$

For independent reads the covariance is 0, and we get Step B again.

**Worked example 4e: two reads from a case-B replicate**

Variance of one read: $0.2 \times 0.8 = 0.16$. Covariance: $V = 0.01$.
Variance of the sum: $0.16 + 0.16 + 2 \times 0.01 = 0.34$.
A computer drew 100,000 pairs and found 0.340.

**$N$ reads.** Now square the sum of $N$ distances, $(d_1 + d_2 + \cdots + d_N)^2$. Multiply every part by every part. You get two kinds of products:

- **Squares** $d_1 \times d_1$, $d_2 \times d_2$, ...: there are $N$ of them. Each has the average $\mu \times (1 - \mu)$.
- **Pairs** $d_1 \times d_2$, $d_2 \times d_1$, ...: each read pairs with each of the $N - 1$ other reads. So there are $N \times (N - 1)$ of them. Each has the average covariance $V$.

Example with $N = 3$ reads: the pairs are 1–2, 1–3, 2–1, 2–3, 3–1, 3–2. That is $6 = 3 \times 2$ pairs. (With two reads, there are $2 \times 1 = 2$ pairs. That is the "2×" above.)

So

$$\text{variance} = \underbrace{N \times \mu \times (1-\mu)}_{N \text{ squares}} + \underbrace{N \times (N-1) \times V}_{N\times(N-1) \text{ pairs}}.$$

**This is where $N \times (N-1)$ comes from: it is the number of pairs of reads.** Every pair of reads from the same replicate adds its covariance $V$.

**The final form.** Put in $V = \rho \times \mu \times (1-\mu)$ from Part 3:

$$\text{variance} = N \times \mu \times (1-\mu) + N \times (N-1) \times \rho \times \mu \times (1-\mu).$$

Both parts contain $N \times \mu \times (1-\mu)$. Take it out, as in $a + a \times b = a \times (1+b)$:

$$\boxed{\text{variance} = \underbrace{N \times \mu \times (1-\mu)}_{\text{counting noise}} \times \underbrace{\big(1 + (N-1) \times \rho\big)}_{\text{extra factor from biological noise}}}$$

Figure 3 shows the whole path.

| **Step** | **Formula** / *check with case B, $N = 10$* |
|---|---|
| 1. The wobble | $V$ = average of $(p-\mu)^2$ |
| | $V = 0.01$ |
| 2. Covariance of two reads | covariance = (both are our guide) $-\mu \times \mu = V$ |
| | $0.05 - 0.04 = 0.01$ |
| 3. Correlation of two reads | $\rho$ = covariance $\div$ (SD $\times$ SD) $= V \div (\mu \times (1-\mu))$ |
| | $0.01 \div 0.16 = 0.0625$ |
| 4. Turn it around | $V = \rho \times \mu \times (1-\mu)$ |
| | $0.0625 \times 0.16 = 0.01$ |
| 5. Add up $N$ reads | $N \times \mu \times (1-\mu) + N \times (N-1) \times V$ |
| | $1.6 + 90 \times 0.01 = 2.5$ |
| 6. Replace $V$ by step 4 | $N \times \mu \times (1-\mu) + N \times (N-1) \times \rho \times \mu \times (1-\mu)$ |
| | $1.6 + 90 \times 0.0625 \times 0.16 = 2.5$ |
| 7. Take out the common part | $N \times \mu \times (1-\mu) \times (1 + (N-1) \times \rho)$ |
| | $1.6 \times (1 + 9 \times 0.0625) = 1.6 \times 1.5625 = 2.5$ |

Figure 3: How $\rho$ gets into the variance formula. Steps 1–4 come from Part 3, and step 5 from Part 4. In step 6, the orange $V$ is replaced by $\rho \times \mu \times (1-\mu)$ from step 4. The lines in italics check each step with case B and $N = 10$ reads, where $N \times \mu \times (1-\mu) = 10 \times 0.2 \times 0.8 = 1.6$ and $N \times (N-1) = 90$.

**Worked example 4f: check with ten reads**

Case B ($\mu = 0.2$, $V = 0.01$, $\rho = 0.0625$) with $N = 10$ reads.

- Squares: $10 \times 0.16 = 1.6$.
- Pairs: $10 \times 9 = 90$ pairs, each 0.01: $90 \times 0.01 = 0.9$.
- Total: $1.6 + 0.9 = 2.5$.
- Boxed formula: $1.6 \times (1 + 9 \times 0.0625) = 1.6 \times 1.5625 = 2.5$.

A computer that drew 100,000 such replicates found the variance 2.498. With $N \times N$ in place of $N \times (N - 1)$, the formula would give 2.6, which is wrong: a read does not form a pair with itself.

**Worked example 4g: the total variance of our guide**

- Squares: $1{,}000{,}000 \times 0.000191 \times 0.999809 = 191.0$.
- Pairs: $1{,}000{,}000 \times 999{,}999 \times 0.0000000000737 = 73.7$.
- Total: $191.0 + 73.7 = 264.7$.
- Boxed formula: $191 \times (1 + 999{,}999 \times 0.000000386) = 191 \times 1.386 = 264.7$.

Each pair adds very little, but there are almost a million million pairs.
**Standard deviation:** $\sqrt{264.7} = 16.3$ reads. Counting noise alone gives $\sqrt{191} = 13.8$ reads. Biological noise adds 39% to the variance.

**Remember**

Look at the extra factor $1 + (N - 1) \times \rho$. It contains $N$. When the machine reads more, $N$ grows, and the factor grows too. So more reading does not remove biological noise. Only more replicates help with biological noise.

# Step 5: The design matrix

A **model** is a rule that predicts a value. BARCS predicts the log-odds of a guide in each library. The model needs to know which group each library belongs to. BARCS gets this information from the **design matrix**.

## What a design matrix is

A design matrix is a table of numbers. It has:

- one **row** for each library, and

- one **column** for each $\beta$ value in the model.

We write the design matrix with the letter $X$.

For our four libraries, the model has two $\beta$ values. So $X$ has four rows and two columns:

| Library | Group | Column 1: "intercept" | Column 2: "drug" |
|---|---|---|---|
| 1 | control | 1 | 0 |
| 2 | control | 1 | 0 |
| 3 | drug | 1 | 1 |
| 4 | drug | 1 | 1 |

**What the symbols mean**

| | |
|---|---|
| $X$ | The design matrix. One row for each library, one column for each $\beta$. |
| Intercept | A column of 1s. It belongs to $\beta_0$, the starting level. Every library gets the starting level. |
| Drug | A column that is 1 for a drug library and 0 for a control library. It belongs to $\beta_1$, the effect of the drug. |
| $\beta_0$ | "beta zero". The log-odds of the guide in the control group. |
| $\beta_1$ | "beta one". The change in log-odds from control to drug. This is the **effect**. It is the most important number. |

## From the design matrix to a prediction

To get the prediction for one library, take the row of that library. Multiply each number in the row by its $\beta$. Then add the products:

$$\text{predicted log-odds} = (\text{column 1}) \times \beta_0 + (\text{column 2}) \times \beta_1.$$

This calculation is a **matrix multiplication**. You do it one row at a time. We call the predicted log-odds $\eta$ ("eta").

**Worked example 5: predictions from the design matrix**

BARCS finds $\beta_0 = -8.565$ and $\beta_1 = -0.542$ (Step 6 in Chapter 2 shows how).

Library 1 (row 1, 0): $1 \times (-8.565) + 0 \times (-0.542) = -8.565$

Library 2 (row 1, 0): $1 \times (-8.565) + 0 \times (-0.542) = -8.565$

Library 3 (row 1, 1): $1 \times (-8.565) + 1 \times (-0.542) = -9.107$

Library 4 (row 1, 1): $1 \times (-8.565) + 1 \times (-0.542) = -9.107$

The control libraries get $\beta_0$ only. The drug libraries get $\beta_0 + \beta_1$. So $\beta_1$ is the difference between the groups.

## From log-odds back to counts

A prediction in log-odds is hard to read. So we change it back into a proportion, and then into a count. The rule is:

$$\mu = \frac{e^{\eta}}{1 + e^{\eta}}, \qquad \text{expected count} = N \times \mu.$$

This rule undoes the log-odds of Step 3.

**Worked example 6: expected counts**

**Control libraries** ($\eta = -8.565$): $e^{-8.565} = 0.000190676$. Then $\mu = 0.000190676 \div 1.000190676 = 0.000190640$, which is 190.6 per million.
**Drug libraries** ($\eta = -9.107$): $e^{-9.107} = 0.000110840$. Then $\mu = 0.000110827$, which is 110.8 per million.

| Library | $N$ | $\mu$ (per million) | Expected count $N \times \mu$ | Real count $K$ |
|---|---|---|---|---|
| 1 | 1,000,000 | 190.6 | 190.6 | 180 |
| 2 | 1,200,000 | 190.6 | 228.8 | 240 |
| 3 | 1,000,000 | 110.8 | 110.8 | 120 |
| 4 | 800,000 | 110.8 | 88.7 | 80 |

The expected counts are near the real counts. The differences are $-10.6$, $+11.2$, $+9.2$ and $-8.7$ reads. These differences are the noise.

## The reference group

The control group has 0 in the drug column. So control is the **reference group**. The effect $\beta_1$ always compares a group with the reference group. If you make drug the reference group, $\beta_1$ changes sign. It becomes $+0.542$. The meaning stays the same.

## Paired replicates: add a column

Sometimes the replicates are pairs. For example, replicate 1 of control and replicate 1 of drug come from the same batch of cells. Then BARCS can remove the batch difference. You add a column for the batch:

| Library | Group | Replicate | Intercept | Replicate 2 | Drug |
|---|---|---|---|---|---|
| 1 | control | 1 | 1 | 0 | 0 |
| 2 | control | 2 | 1 | 1 | 0 |
| 3 | drug | 1 | 1 | 0 | 1 |
| 4 | drug | 2 | 1 | 1 | 1 |

Now the model has three $\beta$ values: the starting level, the batch difference and the drug effect. Each new column costs one degree of freedom (Step 8 in Chapter 2). With four libraries and three columns, only one degree of freedom stays.

**Worked example 7: the paired model for our guide**

BARCS gives the drug effect $\beta = -0.544$ with standard error 0.143. The effect is almost the same as before ($-0.542$). But the degrees of freedom drop from 2 to 1. The p-value becomes 0.164 instead of 0.033.
Our two replicates have almost no batch difference ($\beta = 0.005$). So the extra column costs information and gives nothing back. Add a batch column only when the replicates really are pairs.

## Numbers instead of groups

A column can also hold a number, for example the day of the experiment. Suppose that you collect cells on day 0, day 7 and day 14, two libraries each. You put the day divided by 14 into a column:

| Library | Day | Intercept | Day ÷ 14 |
|---|---|---|---|
| 1, 2 | 0 | 1 | 0 |
| 3, 4 | 7 | 1 | 0.5 |
| 5, 6 | 14 | 1 | 1 |

Now $\beta_1$ is the change in log-odds over 14 days. The model draws a straight line through the three days.

**Worked example 8: a time course**

Suppose that $\beta_1 = -0.70$.
After 14 days: change $= 1 \times (-0.70) = -0.70$. Fold change $= e^{-0.70} = 0.50$. The guide drops to half.
After 7 days: change $= 0.5 \times (-0.70) = -0.35$. Fold change $= e^{-0.35} = 0.70$. The guide drops to 70%.
The model has 6 libraries and 2 columns. So it has $6 - 2 = 4$ degrees of freedom.

## Two factors and an interaction

Some experiments change two things. For example, cells can be normal or can have a second gene switched off (“knockout”). And cells can get no drug or the drug. This experiment has four groups. The design matrix has four columns:

| Group | Intercept | Knockout | Drug | Knockout × Drug |
|---|---|---|---|---|
| normal, no drug | 1 | 0 | 0 | 0 |
| knockout, no drug | 1 | 1 | 0 | 0 |
| normal, drug | 1 | 0 | 1 | 0 |
| knockout, drug | 1 | 1 | 1 | 1 |

The last column is the product of the knockout column and the drug column. It is 1 only when both are 1. Its $\beta$ is the **interaction**. The interaction measures the extra change that happens only when both changes happen together.

**Worked example 9: reading an interaction**

Suppose that the four $\beta$ values are $-8.5$, $-0.2$, $+0.1$ and $-0.7$. Multiply each row by the $\beta$ values and add:

normal, no drug: $-8.5$

knockout, no drug: $-8.5 + (-0.2) = -8.7$

normal, drug: $-8.5 + 0.1 = -8.4$

knockout, drug: $-8.5 + (-0.2) + 0.1 + (-0.7) = -9.3$

The knockout alone changes the log-odds by $-0.2$. The drug alone changes it by $+0.1$. If the two effects only added up, the last group would be $-8.5 - 0.2 + 0.1 = -8.6$. The real value is $-9.3$. The extra $-0.7$ is the interaction. The guide drops strongly only when both changes happen.

## Comparing two treatments: a contrast

Suppose an experiment has three groups: control, drug A and drug B. The design matrix has the columns intercept, drug A and drug B. Then $\beta_A$ compares drug A with control, and $\beta_B$ compares drug B with control. To compare drug A with drug B, you subtract: $\beta_A - \beta_B$. This calculation is a **contrast**. In BARCS, the function `bb_contrast()` calculates it and its test.

For example, if $\beta_A = -0.9$ and $\beta_B = -0.4$, then $\beta_A - \beta_B = -0.5$. Drug A lowers the guide more than drug B.

## Rules for a good design matrix

1. Make one row for each library, in the same order as the count table.
2. Make one column for each $\beta$ that you want.
3. Keep more rows than columns. The difference is the degrees of freedom. Two or more is better than one.
4. Do not let two columns carry the same information. For example, suppose that all drug libraries come from batch 2. Suppose also that all control libraries come from batch 1.

Then the batch column and the drug column are equal. BARCS cannot separate the drug effect from the batch effect.

### The design matrix in R

In R, you describe the design matrix with a **formula**. The formula `~ group` means "an intercept and a column for each group except the reference group".

The function `model.matrix()` builds $X$ from the formula:

```
> data <- data.frame(group = c("control", "control", "drug", "drug"))
> model.matrix(~ group, data)
  (Intercept) groupdrug
1           1         0
2           1         0
3           1         1
4           1         1
```

Then you give the counts, the totals, the formula and the column name to BARCS:

```
fit <- bb_screen(counts, data, formula = ~ group,
                 term = "groupdrug", totals = totals)
```

For the paired design, use the formula `~ replicate + group`. For the interaction, use `~ knockout * drug`.

## Chapter 1 summary

Chapter 1 showed what BARCS measures and how it describes an experiment:

1. It turns read counts into proportions (Step 1).
2. It compares groups with a fold change and its logarithm (Step 2).
3. It works on the log-odds scale (Step 3).
4. It separates counting noise from biological noise (Step 4).
5. It describes the experiment with a design matrix (Step 5).

Chapter 2 shows how BARCS calculates the effect and decides if it is real.

## The symbols of Chapter 1

| Symbol | Meaning | Value in our example |
|---|---|---|
| $K$ | read count of the guide in one library | 180, 240, 120, 80 |
| $N$ | total read count of one library | 1.0, 1.2, 1.0, 0.8 million |
| $p$ | proportion, $K/N$ | 191 and 111 per million |
| fold change | $p_{\text{drug}} \div p_{\text{control}}$ | 0.58 |
| $\log_2$ | number of doublings | $\log_2(0.58) = -0.78$ |
| ln | natural logarithm (powers of $e = 2.718$) | $\ln(0.58) = -0.54$ |
| log-odds | $\ln(p/(1-p))$ | $-8.56$ (control) |
| $\mu$ | expected proportion | 0.000191 (control) |
| $\rho$ | biological noise | 0.000000386 |
| $X$ | design matrix: a row per library, a column per $\beta$ | 4 rows, 2 columns |
| $\beta_0$ | control log-odds in the model | $-8.565$ |
| $\beta_1$ | effect of the drug | $-0.542$ |
| $\eta$ | predicted log-odds of a library (row of $X$ times $\beta$) | $-8.565$, $-9.107$ |

# Exercises for Chapter 1

These exercises use a new screen. Work on paper. Show each calculation, and write one or two sentences when a question asks you to explain. The answers are in a separate answer sheet.

The tags tell you what to do: [Calculate] work out a number, [Explain] say why in your own words, [Write] write down a table or a matrix, [Decide] choose and give a reason.

**The screen.** A lab tests a new drug. They grow cells with a guide library, split them into four flasks, and treat two flasks with the drug. After two weeks they read the guides. Here are the results for one guide, **guide g7**:

| Library | Group | Guide count $K$ | Total count $N$ |
|---|---|---|---|
| 1 | control | 150 | 1,000,000 |
| 2 | control | 210 | 1,400,000 |
| 3 | drug | 60 | 800,000 |
| 4 | drug | 45 | 600,000 |

## Part A: From counts to a change

**1.** [Calculate]

(a) Calculate the proportion of guide g7 in each library, in reads per million.

(b) Calculate the combined proportion of each group (add the counts, add the totals, then divide).

(c) Calculate the fold change (drug ÷ control) and the $\log_2$ fold change.

**2.** [Explain] A student looks only at the counts. Library 2 has 210 reads and library 3 has 60. The student says: "The drug cut guide g7 by $210 \div 60 = 3.5$ times." What is wrong with this? Why must we divide by $N$ before we compare libraries?

**3.** [Calculate] [Explain]

(a) Calculate the log-odds of the control proportion, $\ln\big(0.00015/(1-0.00015)\big)$.

(b) BARCS works with log-odds, not with proportions. Give two reasons. (Hint: what values can a proportion have, and what values can a log-odds have? What happens to "half" and "double" on the log scale?)

## Part B: Noise

**4.** [Calculate] [Explain] Four repeated counts are 140, 150, 160 and 150. Their average is 150.

(a) Calculate the distance of each count from 150, and add the four distances.

(b) Calculate the average squared distance (the variance) and its square root (the standard deviation).

(c) Use part (a) to explain why the variance uses *squared* distances.

**5.** [Calculate] Use library 1: $N = 1{,}000{,}000$ and the expected proportion $\mu = 0.00015$.

(a) Calculate the counting noise $N \times \mu \times (1 - \mu)$ and its standard deviation.

(b) BARCS finds $\rho = 0.000001$. Calculate the extra factor $1 + (N - 1) \times \rho$, the total variance, and the standard deviation.

(c) Calculate the biological part $N \times (N - 1) \times \rho \times \mu \times (1 - \mu)$ on its own. Check that counting part + biological part = total.

**6.** [Explain]

(a) In words, what does $\rho = 0.000001$ say about two reads from the same library?

(b) What would $\rho$ be if every flask had exactly the same true proportion of guide g7?

**7.** [Calculate] [Explain] A tiny library has $N = 4$ reads, called A, B, C and D.

(a) List every ordered pair of two different reads (A–B and B–A are different pairs). How many are there?

(b) Check your count with the formula $N \times (N - 1)$.

(c) Why is the formula $N \times (N - 1)$ and not $N \times N$?

**8.** [Decide] The lab can pay for 8 million reads in each group. Keep $\mu = 0.00015$ and $\rho = 0.000001$. They have three options:

| Option | Replicates per group | Reads per library |
|---|---|---|
| A | 2 | 4,000,000 |
| B | 4 | 2,000,000 |
| C | 8 | 1,000,000 |

(a) For one library, the standard deviation of the proportion is $\sqrt{\text{variance of } K} \div N$. Calculate it (in reads per million) for $N = 4$, 2 and 1 million.

(b) The average of $n$ replicates has the standard deviation of one library divided by $\sqrt{n}$. Calculate it for A, B and C.

(c) Which option measures the group proportion most precisely? Explain why, using the extra factor $1 + (N - 1) \times \rho$.

## Part C: The design matrix

**9.** [Write] A second experiment has 3 control libraries and 3 drug libraries. Write the design matrix $X$ with the columns "intercept" and "drug". How many degrees of freedom does the model have?

**10.** [Write] [Explain] A third experiment uses cells from two donors. Each donor gives one control library and one drug library.

(a) Write $X$ with the columns "intercept", "donor 2" and "drug". Put the libraries in this order: donor 1 control, donor 1 drug, donor 2 control, donor 2 drug.

(b) What is the job of the "donor 2" column? What could go wrong without it?

(c) How many degrees of freedom are left? What does this mean for the test?

**11.** [Calculate] For guide g7, BARCS finds $\beta_0 = -8.8048$ and $\beta_1 = -0.6931$.

(a) Use the drug row of $X$, $(1, 1)$, to calculate the predicted log-odds $\eta$ of a drug library.

(b) Change $\eta$ into the proportion $\mu = e^{\eta}/(1 + e^{\eta})$. Give it in reads per million.

(c) Calculate the expected count of library 3 ($N = 800{,}000$). Compare it with the real count.

**12.** [Explain] Calculate $e^{\beta_1}$. What does this number mean for the experiment? Why is it useful that $\beta_1$ is on the log-odds scale?

# Chapter 2

## The calculations

How BARCS fits the model and decides if an effect is real

# Step 6: How BARCS finds the best $\beta$ values

BARCS cannot find $\beta_0$ and $\beta_1$ in one calculation. It uses a loop of "guess, check, improve". The name of this method is **iteratively reweighted least squares** (IRLS). Each pass through the loop is an **iteration**.

## The loop

1. **Guess.** Start with the simple log-odds of each group (Step 3 in Chapter 1).
2. **Predict.** Use the design matrix to predict $\eta$ and $\mu$ for each library (Step 5 in Chapter 1).
3. **Measure the noise.** Find $\rho$ from the differences between the real and the expected counts (Step 7).
4. **Weigh.** Give each library a **weight**. A library with less noise gets a larger weight.
5. **Correct.** Calculate a corrected value, the **working value**, for each library.
6. **Improve.** Take a weighted average of the working values in each group. These averages give the new $\beta$ values.
7. **Repeat** from part 2 until the $\beta$ values stop changing.

## The working value: a straight-line step

BARCS works on the log-odds scale. The real log-odds of library $i$ is $\ln\Big((K_i/N_i)/(1 - K_i/N_i)\Big)$. But BARCS does not use this value directly. If $K_i = 0$, the value is $\ln(0)$, which does not exist. So BARCS uses a straight line instead.

**The slope of the log-odds.** Near a proportion $\mu$, the log-odds changes by about $1 \div \Big(\mu \times (1 - \mu)\Big)$ for each unit of change in the proportion. Here is why. The log-odds is $\ln(p) - \ln(1 - p)$. Look at the two parts one at a time, with $p = 0.2$ and a small step of 0.001:

| Part | At $p = 0.2$ | At $p = 0.201$ | Change $\div$ 0.001 |
|---|---|---|---|
| $\ln(p)$ | $-1.60944$ | $-1.60445$ | $4.99 \approx 1 \div 0.2 = 5$ |
| $-\ln(1 - p)$ | $+0.22314$ | $+0.22439$ | $1.25 = 1 \div 0.8$ |
| log-odds (sum) | $-1.38629$ | $-1.38006$ | $6.24 \approx 6.25$ |

So the slope is $1 \div p + 1 \div (1 - p)$. Put the two fractions over one bottom:

$$\frac{1}{p} + \frac{1}{1-p} = \frac{(1-p)+p}{p \times (1-p)} = \frac{1}{p \times (1-p)}.$$

Check: $1 \div (0.2 \times 0.8) = 1 \div 0.16 = 6.25$.

**The working value.** Start at the current prediction $\eta_i$ (the log-odds of $\mu_i$). Move along the straight line by the distance $K_i/N_i - \mu_i$:

$$v_i = \eta_i + (K_i/N_i - \mu_i) \times \frac{1}{\mu_i \times (1 - \mu_i)}.$$

The letter $v$ stands for "working value". It is close to the real log-odds, and it exists even when $K_i = 0$.

**Worked example 10a: working value of library 1**

Use the first guess $\mu_1 = 0.000190909$, so $\eta_1 = -8.563523$.

- Distance: $K_1/N_1 - \mu_1 = 0.000180 - 0.000190909 = -0.0000109$.
- Slope: $1 \div (0.000190909 \times 0.999809) = 1 \div 0.000190873 = 5{,}239$.
- Step: $-0.0000109 \times 5{,}239 = -0.0572$.
- Working value: $v_1 = -8.5635 + (-0.0572) = -8.6207$.

The real log-odds is $\ln(0.000180/0.999820) = -8.6224$. The straight line gives $-8.6207$. They are very close.

## The weight: one divided by the variance

**Why weights?** Suppose you have two measurements of the same thing. One has the variance 1. The other has the variance 4 (it is noisier).

- With equal weights, the average has the variance $(1 + 4) \div (2 \times 2) = 1.25$.
- With the weights $1 \div 1 = 1$ and $1 \div 4 = 0.25$, the average has the variance $1 \div (1 + 0.25) = 0.8$. That is smaller.

The rule "weight = 1÷ variance" gives the average with the smallest variance. So BARCS gives each working value the weight

$$w_i = \frac{1}{\text{variance of } v_i}.$$

**The variance of the working value.** We build it from Chapter 1 in three steps. The right column uses library 1 in iteration 1 ($N = 1{,}000{,}000$, $\mu = 0.000190909$, $\rho = 0.0000003841$).

| Quantity | Formula | Library 1 |
|---|---|---|
| 1. variance of $K_i$ | $N_i \times \mu_i \times (1-\mu_i) \times (1+(N_i-1)\times\rho)$ | $190.87 \times 1.3841 = 264.19$ |
| 2. variance of $K_i/N_i$ | line 1 $\div(N_i \times N_i)$ | $264.19 \div 10^{12}$ |
| | | $= 2.642 \times 10^{-10}$ |
| 3. variance of $v_i$ | line 2 $\div(\mu_i \times (1-\mu_i))^2$ | $2.642 \times 10^{-10} \div 0.000190873^2$ |
| | | $= 0.007251$ |
| 4. weight $w_i$ | 1$\div$ line 3 | $1 \div 0.007251 = 137.90$ |

- **Line 1** is the variance formula of Chapter 1, Step 4.
- **Line 2:** dividing a value by $N$ divides its variance by $N \times N$ (Chapter 1, Part 4 used the same rule with multiplying).
- **Line 3:** the working value multiplies $K_i/N_i - \mu_i$ by the slope $1 \div \Big(\mu_i \times (1-\mu_i)\Big)$. So the variance is multiplied by the slope squared. Adding $\eta_i$ does not change the variance.
- **Line 4:** put lines 1–3 together and simplify. One $N_i$ and one $\mu_i \times (1-\mu_i)$ cancel:

$$w_i = \frac{1}{\text{variance of } v_i} = \frac{N_i \times N_i \times \Big(\mu_i \times (1-\mu_i)\Big)^2}{N_i \times \mu_i \times (1-\mu_i) \times \Big(1+(N_i-1)\times\rho\Big)} = \frac{N_i \times \mu_i \times (1-\mu_i)}{1+(N_i-1)\times\rho}.$$

Check: $190.87 \div 1.3841 = 137.90$, the same as line 4.

A library with more reads gets a bigger top, so a bigger weight. More biological noise ($\rho$) gives a bigger bottom, so a smaller weight.

## The new $\beta$ values

For our design matrix, the new $\beta$ values are weighted averages of the working values:

$$\beta_0 = \frac{w_1 \times v_1 + w_2 \times v_2}{w_1 + w_2}, \qquad \beta_0 + \beta_1 = \frac{w_3 \times v_3 + w_4 \times v_4}{w_3 + w_4}.$$

(For other design matrices, BARCS solves a small set of equations instead. The idea is the same.)

**Why a loop?** The weights and the working values use $\mu$. But $\mu$ comes from the $\beta$ values, and the $\beta$ values come from the weights. So BARCS starts with a guess and repeats until nothing changes.

## Iteration 1 in full

**Worked example 10b: iteration 1**

**Guess.** From Step 3 in Chapter 1, the control log-odds is $-8.563522$ and the drug log-odds is $-9.104869$. So $\beta_0 = -8.563522$ and $\beta_1 = -9.104869 - (-8.563522) = -0.541347$.

**Predict.** The control libraries get $\mu = 0.000190909$. The drug libraries get $\mu = 0.000111111$.

**Measure the noise.** Step 7 gives $\rho = 0.0000003841$ for these predictions.

**Weigh library 1.** Top: $1{,}000{,}000 \times 0.000190909 \times 0.999809 = 190.87$.
Bottom: $1 + 999{,}999 \times 0.0000003841 = 1.3841$.
Weight: $w_1 = 190.87 \div 1.3841 = 137.90$.

**Correct library 1.** Example 10a: $v_1 = -8.6207$.

The same calculations for all four libraries give:

| Library | Weight $w_i$ | Working value $v_i$ | $w_i \times v_i$ |
|---|---|---|---|
| 1 (control) | 137.905 | −8.62068 | −1188.84 |
| 2 (control) | 156.785 | −8.51589 | −1335.16 |
| 3 (drug) | 80.269 | −9.02486 | −724.42 |
| 4 (drug) | 67.988 | −9.20488 | −625.82 |

**Improve.**
Control: $\beta_0 = (-1188.84 - 1335.16) \div (137.905 + 156.785) = -2524.00 \div 294.69 = -8.564929$.
Drug: $\beta_0 + \beta_1 = (-724.42 - 625.82) \div (80.269 + 67.988) = -1350.24 \div 148.257 = -9.107414$.
So $\beta_1 = -9.107414 - (-8.564929) = -0.542486$.

## The next iterations

BARCS repeats the loop with the new values. The changes become very small:

| After iteration | $\beta_0$ | $\beta_1$ |
|---|---|---|
| 0 (the guess) | −8.563522 | −0.541347 |
| 1 | −8.564929 | −0.542486 |
| 2 | −8.564934 | −0.542492 |
| 3 | −8.564934 | −0.542492 |

After iteration 3, the values do not change in the first six decimal places. BARCS stops. The final values are $\beta_0 = -8.565$ and $\beta_1 = -0.542$.

**Remember**

The weighted average gives a slightly different answer from the simple log-odds of Step 3 in Chapter 1 ($-0.5425$ instead of $-0.5413$). The difference comes from the weights. Library 2 has the most reads and the least relative noise. So it counts the most.

# Step 7: How BARCS measures the biological noise

Step 4 in Chapter 1 used $\rho = 0.000000386$. This section shows where the number comes from.

## Pearson residuals

For each library, BARCS subtracts the expected count from the real count. It divides the result by the standard deviation from Step 4 in Chapter 1. The result is a **Pearson residual**:

$$\text{Pearson residual}_i = \frac{K_i - N_i \times \mu_i}{\sqrt{N_i \times \mu_i \times (1 - \mu_i) \times \Big(1 + (N_i - 1) \times \rho\Big)}}.$$

A Pearson residual tells how many standard deviations a library is from its prediction.

## The rule for $\rho$

**Why the sum should be the degrees of freedom.** A Pearson residual is a distance measured in standard deviations. If the noise model is correct, its square is 1 on average: (squared distance) ÷ (variance) = variance ÷ variance = 1. So with four libraries, the four squares add up to about 4.

But BARCS does not know the true $\mu$. It uses the fitted $\mu$, which the $\beta$ values pull toward the data. The fit makes the residuals a little smaller. Each fitted $\beta$ value takes away 1 from the sum, on average. So the expected sum is

$$(\text{number of libraries}) - (\text{number of } \beta \text{ values}) = 4 - 2 = 2 = \text{df}.$$

A computer checked this. It made 20,000 artificial guides with the true $\rho$ and four libraries like ours. With the true $\mu$, the sum of the squared Pearson residuals was 4.00 on average. With the fitted $\mu$, it was 2.00 on average.

**The rule.** BARCS turns this around. It chooses $\rho$ so that the sum is exactly the degrees of freedom:

$$\text{sum of the squared Pearson residuals} = \text{df}.$$

(Statisticians call this a *method of moments*: make a sum that you can calculate equal to the value it should have on average.)

**Worked example 11: start with $\rho = 0$**

Use the expected counts from Step 5 in Chapter 1. With $\rho = 0$, the bottom of each residual is the counting noise only.

| Library | $(K - N \times \mu)^2$ | $N \times \mu \times (1 - \mu)$ | Squared residual |
|---|---|---|---|
| 1 | $(-10.64)^2 = 113.21$ | 190.60 | 0.594 |
| 2 | $(+11.23)^2 = 126.16$ | 228.72 | 0.552 |
| 3 | $(+9.17)^2 = 84.14$ | 110.82 | 0.759 |
| 4 | $(-8.66)^2 = 75.03$ | 88.65 | 0.846 |
| | | Sum | 2.751 |

The sum is 2.751. The degrees of freedom are 2. The sum is too big. So there is more noise than counting noise alone. We need $\rho > 0$.

**Worked example 12: guess and check for $\rho$**

A larger $\rho$ makes the bottom of each residual larger. So the sum goes down. BARCS tries values until the sum equals 2:

| $\rho$ | Sum of squared residuals |
|---|---|
| 0 | 2.751 |
| 0.0000001 | 2.506 |
| 0.0000002 | 2.302 |
| 0.0000003 | 2.129 |
| 0.000000386 | 2.000 |
| 0.0000005 | 1.851 |

The answer is $\rho = 0.000000386$. If the sum is already 2 or less at $\rho = 0$, BARCS uses $\rho = 0$.

The ratio $2.751 \div 2 = 1.376$ is also useful. It is the **variance inflation**. At $\rho = 0$, the sum should be about 2; it is 2.751. So the variance is about 1.38 times bigger than counting noise alone. This is close to the extra factor $1 + (N - 1) \times \rho = 1.386$ of Chapter 1. Step 9 uses the inflation.

# Step 8: Is the effect real?

The effect is $\beta_1 = -0.542$. But replicates are noisy. Maybe the true effect is 0 and the noise made the drop. BARCS tests this question in four parts.

## The standard error

The **standard error** (SE) tells how much the estimate of $\beta_1$ can change because of noise. A small SE means a precise estimate.

For our design matrix, the standard error comes from the weights of Step 6. Add the weights of each group:

$$W_{\text{control}} = w_1 + w_2, \qquad W_{\text{drug}} = w_3 + w_4.$$

Then

$$\text{SE}(\beta_1) = \sqrt{\frac{1}{W_{\text{control}}} + \frac{1}{W_{\text{drug}}}}.$$

**Where this formula comes from.** It uses only three facts: the weight is $1 \div$ variance (Step 6); variances of independent values add up; and multiplying a value by $c$ multiplies its variance by $c \times c$.

| **Step** | **Why** |
|---|---|
| 1. variance of $v_i = 1 \div w_i$ | definition of the weight |
| 2. $\beta_0 = \dfrac{w_1}{W_c} \times v_1 + \dfrac{w_2}{W_c} \times v_2 \quad (W_c = w_1 + w_2)$ | Step 6 |
| 3. variance of $\beta_0 = \left(\dfrac{w_1}{W_c}\right)^2 \times \dfrac{1}{w_1} + \left(\dfrac{w_2}{W_c}\right)^2 \times \dfrac{1}{w_2}$ | multiply by $c \Rightarrow$ variance $\times\, c \times c$; add |
| 4. $= \dfrac{w_1 + w_2}{W_c \times W_c} = \dfrac{1}{W_c}$ | one $w$ cancels in each part |
| 5. variance of $(\beta_0 + \beta_1) = 1 \div W_{\text{drug}}$ | the same for the drug libraries |
| 6. variance of $\beta_1 = \dfrac{1}{W_c} + \dfrac{1}{W_{\text{drug}}}$ | $\beta_1 = (\beta_0 + \beta_1) - \beta_0$; variances add |
| 7. $\text{SE}(\beta_1) = \sqrt{\text{line 6}}$ | SE is the standard deviation of the estimate |

Line 6 adds the variances even though it subtracts the values. The reason: $-1 \times -1 = +1$, so a minus sign does not change a variance.

More weight means less noise, so the SE becomes smaller. (For other design matrices, BARCS uses the inverse of the matrix $X^\top W X$. For our design, that calculation gives exactly this formula.)

**Worked example 13: the standard error**

The final weights are 137.524, 156.322, 79.955 and 67.737.
$W_{\text{control}} = 137.524 + 156.322 = 293.846$.
$W_{\text{drug}} = 79.955 + 67.737 = 147.692$.
$1 \div 293.846 = 0.003403$ (line 4: $(137.524 + 156.322) \div 293.846^2$ gives the same).
$1 \div 147.692 = 0.006771$.
Sum: $0.003403 + 0.006771 = 0.010174$.
$\text{SE} = \sqrt{0.010174} = 0.101$.
The control group has about twice the weight of the drug group. So most of the uncertainty comes from the drug group.

## The t value

The **t value** compares the effect with its standard error:

$$t = \frac{\beta_1}{\text{SE}}.$$

A $t$ value far from 0 means that the effect is large compared with the noise.

## Degrees of freedom

The **degrees of freedom** (df) tell how much information we have about the noise:

$$\text{df} = (\text{rows of } X) - (\text{columns of } X) = (\text{number of libraries}) - (\text{number of } \beta \text{ values}).$$

Fewer degrees of freedom mean less information. Then BARCS needs a bigger $t$ value before it accepts an effect as real. (The reason is that the SE itself is estimated from few libraries, so it can be too small by chance. W. S. Gosset, who wrote as "Student", worked out the table for this case in 1908.[5])

| Degrees of freedom | 1 | 2 | 14 | very many |
|---|---|---|---|---|
| $\lvert t \rvert$ needed for $p < 0.05$ | 12.71 | 4.30 | 2.14 | 1.96 |

## The p-value

The **p-value** answers this question: "If the true effect is 0, how often does noise alone give a $t$ value this far from 0?" A small p-value means that noise alone seldom gives such a result. Scientists often use 0.05 as a limit.

[5] Student (1908). The probable error of a mean. *Biometrika* 6: 1–25.

**Worked example 14: testing the effect**

1. $t = -0.542 \div 0.101 = -5.38$.
2. $X$ has 4 rows and 2 columns. So df $= 4 - 2 = 2$.
3. The table shows that 2 degrees of freedom need $|t| > 4.30$. Our $|t| = 5.38$. It is larger than 4.30. So $p < 0.05$.
4. A computer gives the exact p-value: $p = 0.033$.

The p-value is less than 0.05. So the drop is probably not only noise.

**Remember**

The degrees of freedom come from the number of libraries, not from the number of reads. A library with 100 million reads still counts as one row of $X$.

## Step 9: Borrow information from other guides

With only 2 degrees of freedom, the noise estimate of one guide is not reliable. But a screen has thousands of guides. Most guides have a similar amount of noise. So BARCS combines two estimates:

- the variance inflation of this guide (Step 7), and
- the typical variance inflation of guides with similar counts.

This method is **moderation**.[6] BARCS takes a weighted average. Each estimate gets a number of "votes". The guide's own estimate gets its degrees of freedom as votes. The typical estimate gets the **prior degrees of freedom**, $d_0$. BARCS calculates $d_0$ and the typical value from all the guides in the screen.

$$\text{moderated inflation} = \frac{\text{df} \times \text{own inflation} + d_0 \times \text{typical inflation}}{\text{df} + d_0}.$$

[6] BARCS uses the empirical-Bayes method of G. K. Smyth (2004). Linear models and empirical Bayes methods for assessing differential expression in microarray experiments. *Statistical Applications in Genetics and Molecular Biology* 3: Article 3.

**Worked example 15: moderation with simple numbers**

A guide has 3 degrees of freedom. Its own inflation is 4.0. The typical inflation is 1.5, with $d_0 = 12$ votes.

$$\text{moderated inflation} = \frac{3 \times 4.0 + 12 \times 1.5}{3 + 12} = \frac{12 + 18}{15} = \frac{30}{15} = 2.0.$$

The guide's value moves from 4.0 toward 1.5. It stops at 2.0.

### What moderation does to the test

BARCS uses the moderated inflation in three places:

1. It multiplies the standard error by $\sqrt{\text{moderated inflation} \div \text{own inflation}}$. Here is why. The variance of $\beta_1$ grows in proportion to the inflation. To swap the own inflation for the moderated one, multiply the variance by (moderated ÷ own). The SE is the square root of the variance, so it is multiplied by the square root. (If the own inflation is below 1, BARCS uses 1, so moderation never makes a test less careful than counting noise alone.)
2. It recalculates $t$ with the new standard error.
3. It adds the $d_0$ votes to the degrees of freedom. So the test uses $\text{df} + d_0$.

**Worked example 16: moderation for our guide**

Our guide has own inflation 1.376 (Step 7) and 2 degrees of freedom. Suppose that the screen gives typical inflation 1.5 with $d_0 = 12$.
Moderated inflation: $(2 \times 1.376 + 12 \times 1.5) \div (2 + 12) = (2.751 + 18) \div 14 = 1.482$.
Factor for the SE: $\sqrt{1.482 \div 1.376} = \sqrt{1.077} = 1.038$.
New SE: $0.101 \times 1.038 = 0.105$.
New $t$: $-0.542 \div 0.105 = -5.18$.
New degrees of freedom: $2 + 12 = 14$.
With 14 degrees of freedom, $|t|$ must be larger than 2.14 for $p < 0.05$. Our $|t| = 5.18$.
The new p-value is 0.00014.
The SE became slightly larger, because the typical guide is a little noisier than our guide. But the degrees of freedom grew from 2 to 14. So the test became much more powerful.

BARCS uses moderation when it has 50 or more usable guides.

## Step 10: Check the results with control guides

A good screen contains **non-targeting control guides**. These guides switch off no gene. So their true effect is 0.

If the noise model is correct, about 5% of control guides get $p < 0.05$ by chance. If many more control guides get $p < 0.05$, the noise model is too small. Then BARCS makes all the tests wider.

**Worked example 17: calibration with control guides**

A screen has 100 control guides. We expect about 5 of them to get $p < 0.05$. But 15 of them get $p < 0.05$. The noise is too small.
BARCS finds a **scale factor** that makes the control guides behave correctly. It sorts the 100 control guides by the size of $|t|$ (the value without its sign). It takes the value that 95 of the 100 guides stay below. Suppose that this value is 2.47. With many degrees of freedom, the $t$ table says that this value should be 1.96. So the scale factor is $2.47 \div 1.96 = 1.26$. BARCS divides every $t$ value by 1.26.
A gene with $t = 3.5$ now gets $t = 3.5 \div 1.26 = 2.78$. Its p-value becomes larger. This change protects you from false results.

The scale factor is never less than 1. So calibration can only make the results more careful.

# Step 11: From guides to genes

Each gene usually has several guides, often 3 to 6. Each guide gives its own result. BARCS combines them into one result for the gene. It uses the **Stouffer method** in three parts.

1. Change each guide's p-value into a **z score**. A z score tells how many standard deviations a result is from 0. The sign of the z score follows the sign of the effect.
2. Add the z scores.
3. Divide the sum by the square root of the number of guides, $m$.

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m}}.$$

**Why divide by $\sqrt{m}$?** If the gene has no effect, each $z$ score has the variance 1. The guides are independent, so the variance of the sum is $1 + 1 + \cdots + 1 = m$ (Chapter 1, Part 2: variances add). Dividing a value by $\sqrt{m}$ divides its variance by $\sqrt{m} \times \sqrt{m} = m$. So $Z_{\text{gene}}$ has the variance $m \div m = 1$ again, and the same z table works for it.[7]

To change a p-value into a z score, use the table of the normal distribution. A two-sided p-value of 0.05 belongs to $z = 1.96$. Some other pairs:

| Two-sided p-value | 0.317 | 0.134 | 0.046 | 0.012 | 0.0027 |
|---|---|---|---|---|---|
| $z$ score | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 |

[7] S. A. Stouffer and others (1949). *The American Soldier*, Vol. 1. Princeton University Press.

If the effect is negative (the guide drops), the z score gets a minus sign.

**Worked example 18: combining three guides**

A gene has $m = 3$ guides. Their p-values are 0.046, 0.134 and 0.012. Their z scores are 2.0, 1.5 and 2.5.

$$Z_{\text{gene}} = \frac{2.0 + 1.5 + 2.5}{\sqrt{3}} = \frac{6.0}{1.732} = 3.46.$$

This $Z$ gives $p = 0.00053$ for the gene. No single guide is that strong. Together, the three guides give strong evidence.

### When guides share noise

Sometimes the guides of one gene share noise. Then they agree more than they should by chance. BARCS measures this sharing with the **guide correlation**, $r$. Then the variance of the sum is not $m$. Use the squares-and-pairs rule from Chapter 1, Part 4:

- $m$ squares, each with the variance 1: together $m$;
- $m \times (m - 1)$ pairs, each with the covariance $r$: together $m \times (m - 1) \times r$.

So BARCS divides by the square root of this larger variance:

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m + m \times (m - 1) \times r}}.$$

**Worked example 19: shared noise**

Use the same three guides and $r = 0.1$. The bottom becomes $\sqrt{3 + 3 \times 2 \times 0.1} = \sqrt{3.6} = 1.897$. So $Z_{\text{gene}} = 6.0 \div 1.897 = 3.16$, and $p = 0.0016$. The evidence is still strong, but less strong than before.

## Step 12: Many genes and false discoveries

A screen tests about 20,000 genes. Suppose that no gene has an effect. If we use the limit $p < 0.05$, about 5% of the genes still pass by chance. That is about 1,000 false results.

So BARCS uses the **false discovery rate** (FDR). The FDR is the expected fraction of false results in your list of hits.

**The idea.** When a gene has no effect, its p-value is equally likely to be anywhere from 0 to 1. So a fraction $p$ of such genes get a p-value of $p$ or less by chance. Example: of 20,000

genes with no effect, about $20{,}000 \times 0.001 = 20$ get $p \leq 0.001$. Now suppose that your list "all genes with $p \leq 0.001$" has 100 genes. At most about 20 of them are there by chance. So the false fraction is about $20 \div 100 = 0.2$. In general:

$$\text{FDR of a list} \approx \frac{p \times (\text{number of genes})}{\text{number of genes in the list}}.$$

If you sort the genes by p-value, the list that ends at a gene has "rank" genes in it. That gives part 3 below.[8]

BARCS calculates it with the **Benjamini–Hochberg method**:

1. Sort the genes from the smallest p-value to the largest.
2. Give each gene a rank: 1, 2, 3, and so on.
3. For each gene, calculate $p \times$ (number of genes) $\div$ rank.
4. Go from the bottom of the list to the top. If a value is larger than the value below it, replace it with the value below it. (A gene is in every longer list too. If a longer list has a smaller FDR, the gene can use that list.)

**Worked example 20: false discovery rate for five genes**

| Gene | Rank | $p$ | $p \times 5 \div$ rank | FDR (after part 4) |
|---|---|---|---|---|
| A | 1 | 0.001 | $0.001 \times 5 \div 1 = 0.005$ | 0.005 |
| B | 2 | 0.008 | $0.008 \times 5 \div 2 = 0.020$ | 0.020 |
| C | 3 | 0.039 | $0.039 \times 5 \div 3 = 0.065$ | 0.051 (from D) |
| D | 4 | 0.041 | $0.041 \times 5 \div 4 = 0.051$ | 0.051 |
| E | 5 | 0.600 | $0.600 \times 5 \div 5 = 0.600$ | 0.600 |

With the limit FDR $< 0.05$, genes A and B are hits. With the limit FDR $< 0.10$, genes A, B, C and D are hits.

# Two special cases

## A guide with zero reads

Sometimes a guide has 0 reads in all control libraries. Then the log-odds of the control group is $\ln(0)$. This value does not exist, because no power of $e$ equals 0. Older versions of BARCS could not fit such a guide. So they removed it.

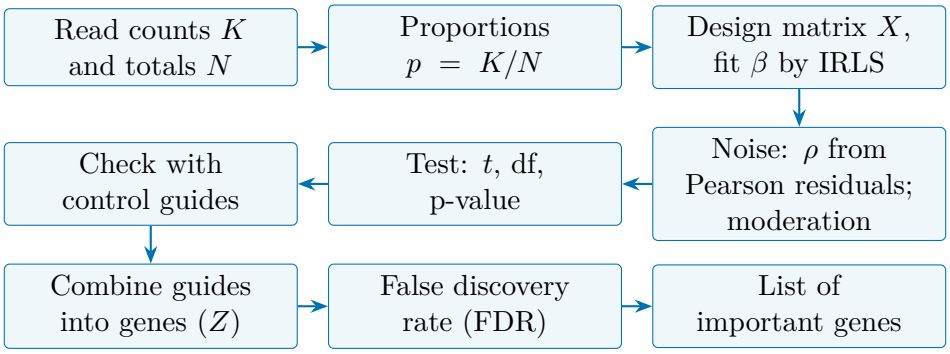
BARCS version 0.2.1 adds a very small number to the counts first. This number is less than one read. The method is the **Firth correction**. Now the guide gets a value, and BARCS keeps it.

[8] Y. Benjamini and Y. Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B* 57: 289–300.

### A very large change

Sometimes a guide grows by hundreds of times. The usual test from Step 8 can then fail. The standard error grows faster than the effect, so the $t$ value becomes small. BARCS version 0.2.1 has a second test, the **likelihood-ratio test**. It compares how well the model fits with the effect and without the effect. The option `test = "auto"` uses this test only for guides with very large changes.

## Summary: the full path



## All symbols in one table

| Symbol | Meaning | Value in our example |
|---|---|---|
| $K$ | read count of the guide in one library | 180, 240, 120, 80 |
| $N$ | total read count of one library | 1.0, 1.2, 1.0, 0.8 million |
| $p$ | proportion, $K/N$ | 191 and 111 per million |
| fold change | $p_{\text{drug}} \div p_{\text{control}}$ | 0.58 |
| $\log_2$ | number of doublings | $\log_2(0.58) = -0.78$ |
| ln | natural logarithm (powers of $e = 2.718$) | $\ln(0.58) = -0.54$ |
| log-odds | $\ln(p/(1-p))$ | $-8.56$ (control) |
| $\mu$ | expected proportion | 0.000191 (control) |
| $\rho$ | correlation of two reads from one library (biological noise) | 0.000000386 |
| $X$ | design matrix: a row per library, a column per $\beta$ | 4 rows, 2 columns |
| $\beta_0$ | control log-odds in the model | $-8.565$ |
| $\beta_1$ | effect of the drug | $-0.542$ |
| $\eta$ | predicted log-odds of a library (row of $X$ times $\beta$) | $-8.565$, $-9.107$ |
| $w$ | weight of a library in the fit | 137.5 (library 1) |
| working value | prediction plus a straight-line step (Example 10a) | $-8.621$ (library 1) |
| Pearson residual | distance from the prediction in SDs | $-0.65$ (library 1) |
| inflation | noise compared with counting noise | 1.376 |
| SE | standard error of the effect | 0.101 |
| $t$ | effect divided by SE | $-5.38$ |
| df | libraries minus $\beta$ values | $4 - 2 = 2$ |
| $p$-value | chance of this result if the effect is 0 | 0.033 |
| $d_0$ | prior degrees of freedom (votes) | 12 (Example 15) |
| $z$, $Z$ | z score of a guide, of a gene | 3.46 (Example 18) |
| $m$ | number of guides for a gene | 3 |
| $r$ | guide correlation (shared noise) | 0.1 (Example 19) |
| FDR | expected fraction of false hits | 0.005 for gene A |

Note: the letter $p$ has two uses in this guide. Alone, $p$ is a proportion. In the word "p-value", it is a probability. The text always tells you which one it is.

# Exercises for Chapter 2

These exercises continue the screen of the Chapter 1 exercises (guide g7). Work on paper and show each calculation. The answers are in a separate answer sheet.

The tags tell you what to do: [Calculate] work out a number, [Explain] say why in your own words, [Decide] choose and give a reason.

**Values to use.** Expected proportions: $\mu = 0.00015$ (control) and $\mu = 0.000075$ (drug). Biological noise: $\rho = 0.000001$. Effect: $\beta_1 = -0.69$. Use ln for the natural logarithm.

## Part A: Fitting

**1.** [Calculate] [Explain] A control library has $K/N = 0.00018$. The current prediction is $\mu = 0.00015$, so $\eta = \ln(0.00015/0.99985) = -8.8047$.

(a) Calculate the slope $1 \div \big(\mu \times (1 - \mu)\big)$.

(b) Calculate the step $(K/N - \mu) \times$ slope and the working value $v = \eta +$ step.

(c) Calculate the real log-odds $\ln(0.00018/0.99982)$ and compare.

(d) Why does BARCS use the straight-line working value and not the real log-odds? Think of a guide with $K = 0$.

**2.** [Calculate] [Explain] Two control libraries have $N = 1{,}000{,}000$ and $N = 2{,}000{,}000$. Use $\mu = 0.00015$.

(a) Calculate the weight $w = N \times \mu \times (1 - \mu) \div \big(1 + (N - 1) \times \rho\big)$ of each library.

(b) The second library has twice the reads. Does it have twice the weight? Explain why not.

(c) What would the two weights be if $\rho = 0$?

**3.** [Calculate] [Explain] Two libraries have the working values $-8.60$ (weight 75) and $-8.90$ (weight 100).

(a) Calculate the weighted average.

(b) Calculate its standard error, $\sqrt{1 \div (75 + 100)}$.

(c) The weighted average is closer to $-8.90$ than to $-8.60$. Why is that the right thing to do?

## Part B: Is the effect real?

**4.** [Calculate] [Explain] The control weights add up to $W_{\text{control}} = 175$. The drug weights add up to $W_{\text{drug}} = 70$.

(a) Calculate $\mathrm{SE}(\beta_1) = \sqrt{1/W_{\text{control}} + 1/W_{\text{drug}}}$.

(b) Calculate $t = \beta_1 \div \mathrm{SE}$ and the degrees of freedom (4 libraries, 2 $\beta$ values).

(c) With 2 degrees of freedom, $|t|$ must be larger than 4.30 for $p < 0.05$. Is the effect significant?

(d) Which group adds more uncertainty? What would you change in the next experiment to make the SE smaller?

**5.** [Calculate] [Explain] Two guides, P and Q, each have 4 libraries and 2 $\beta$ values. At $\rho = 0$, the squared Pearson residuals add up to 5.1 for guide P and to 1.4 for guide Q.

(a) Calculate the variance inflation (sum ÷ df) of each guide.

(b) For which guide does BARCS use $\rho > 0$? For which does it use $\rho = 0$?

(c) Why do we compare the sum with 2 and not with 4?

**6.** [Calculate] [Explain] Guide P has its own inflation 2.55 with 2 degrees of freedom. In the screen, the typical inflation is 1.5 with $d_0 = 10$.

(a) Calculate the moderated inflation.

(b) Calculate the factor for the SE, $\sqrt{\text{moderated} \div \text{own}}$, and the new degrees of freedom.

(c) Why is it reasonable to borrow information from other guides? When could it be a bad idea?

## Part C: From guides to a list of genes

**7.** [Calculate] [Decide] The screen has 300 non-targeting control guides. 45 of them get $p < 0.05$.

(a) How many would you expect if the noise model were correct? What does the real number tell you?

(b) 95% of the control guides have $|t|$ below 2.6. The $t$ table says 1.96. Calculate the scale factor.

(c) A gene has $t = 3.0$. Calculate its new $t$. Is it still beyond 1.96?

**8.** [Calculate] [Explain] A gene has four guides with the z scores 2.2, 1.8, −0.3 and 2.5.

(a) Calculate $Z_{\text{gene}}$ with the Stouffer method (no shared noise).

(b) Calculate $Z_{\text{gene}}$ again with the guide correlation $r = 0.2$.

(c) Why does one guide with a negative $z$ lower the result? Why does shared noise lower the result?

**9.** [Calculate] [Decide] A screen tests 10,000 genes. 40 genes have $p \leq 0.0005$.

(a) If no gene had an effect, about how many genes would have $p \leq 0.0005$ by chance?

(b) Estimate the false discovery rate of this list of 40 genes.

(c) You want a list with an FDR of about 0.05. Should you use a smaller or a larger p-value limit? Explain.

**10.** [Decide] [Explain] A collaborator says: "We will read 20 million reads per library. With so many reads, 2 replicates per group are enough." Use $\rho = 0.000001$ and the degrees of freedom to answer. (Hint: calculate the extra factor $1 + (N-1) \times \rho$ for $N = 20{,}000{,}000$.)